Two-temperature cyclic compression for the Gyarmati-Hennecart-Ruzsa constant

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Status and result

This note gives a new candidate proof that

$$\boxed{C_{3a} > 1.21418}.$$

The construction is not a retuning of the earlier six-generator mask. Its inner alphabet is the much smaller numerical-semigroup truncation

$$M = \langle 4, 5\rangle \cap [0, 80] = \{0, 4, 5, 8, 9, 10\} \cup [12, 80],$$

and its main new ingredient is a **two-temperature entropy theorem**: the difference family and the sum-family upper bound are evaluated at separate dual parameters. The old one-temperature theorem forced those two parameters to coincide.

At the exact rational parameters

$$x = \frac{8149}{10000}, \qquad y = \frac{42571}{50000}, \qquad R = \frac{249293}{10000},$$

the display-only fixed-rate value is

$$1 + \frac{\rho - \sigma}{\log R} = 1.21418089609778549420005709249\ldots,$$

while the entropy-matched limiting value is

$$2 - \frac{\sigma}{\rho} = 1.21418118095053104745496921148\ldots.$$

The conservative claim is only the strict rationally certified inequality $C_{3a} > 1.21418$.

The numerical certificate is machine checked by two independently structured programs. The two-temperature theorem and the finite-set lemma remain written mathematical arguments; this package is not a Lean formalization or an externally refereed publication.

1. The finite-set principle

We use the following lemma from K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici **37** (2007), 175-186, Section 2, pp. 178-179, DOI 10.7169/facm/1229618749.

GHR finite-set lemma. Let $U \subset \mathbb{N}_0$ be finite and nonempty, with $0 \in U$. Put

$$s = |U + U|, \qquad d = |U - U|, \qquad q_U = 2 \max U + 1.$$

If $d < q_U$, then

$$\theta(U) = 1 + \frac{\log(d/s)}{\log q_U}$$

is an admissible exponent for the GHR problem. More precisely, for every $K > 1$ there are arbitrarily large finite integer sets A, B such that $|A + B| \le K|A|$ and

$$|A - B| \ge \left(\frac{2(K-1)}{3K}\right)^{5/4} |A + B|^{\theta(U)}.$$

Thus a single finite U satisfying the strict difference-set condition supplies a lower bound for C_{3a}.

2. Universal cyclic compression

The outer compression step does not depend on the numerical semigroup.

Lemma 2.1 (cyclic quotient with controlled collisions)

Let $V \subset \mathbb{Z}^n$ be finite and contain zero. Let $W \subset V - V$ have $T = |W| \ge 2$, and suppose every coordinate of every word in W has absolute value at most B_0. If p is prime and

$$p > 2B_0, \qquad p > |V - V|,$$

then there is a linear map

$$\phi_c(z) = \sum_{i=1}^{n} c_i z_i \pmod p$$

such that, after multiplying ϕ_c by a nonzero scalar if necessary, the least-nonnegative representative set

$$A = \mathrm{rep}_{[0,p-1]}\, \phi_c(V)$$

satisfies

$$\max A = p - 1, \tag{2.1}$$

$$|A + A| \le 2|V + V|, \tag{2.2}$$

$$|A - A| > \frac{T}{2}, \tag{2.3}$$

and

$$|A - A| \le 2|V - V| - 1 < 2p - 1 = 2\max A + 1. \tag{2.4}$$

The coefficient vector can be chosen canonically by conditional expectations, followed by canonical rescaling.

Proof

Choose all coefficients $c_1, \dots, c_n$ independently and uniformly in $\mathbb{F}_p$. For distinct $w, w' \in W$, write $h = w - w'$. Some coordinate h_j is a nonzero integer with $|h_j| \le 2B_0 < p$, and hence is nonzero in $\mathbb{F}_p$. Holding the other coefficients fixed, exactly one value of c_j solves $\phi_c(h) = 0$. Therefore

$$\Pr(\phi_c(w) = \phi_c(w')) = \frac{1}{p}.$$

If $C(c)$ is the number of unordered colliding pairs in W, then

$$\mathbb{E}C(c) = \frac{\binom{T}{2}}{p} < \frac{T}{2},$$

because $T \le |V - V| < p$. Choose c with $C(c) < T/2$. This choice can be made coefficient by coefficient by minimizing the conditional expected final collision count and breaking ties by the least residue.

If a fiber of $\phi_c|_W$ has size t, it loses $t-1$ image points but contributes $\binom{t}{2} \ge t-1$ colliding pairs. Consequently

$$|\phi_c(W)| \ge T - C(c) > T/2.$$

Since $T \ge 2$, the image $\phi_c(V)$ is not $\{0\}$. Choose the least specified $v \in V$ with $\phi_c(v) \ne 0$, and multiply the map by $-\phi_c(v)^{-1}$. This does not change any cardinality and puts -1 in the image. Since zero is also in the image, the representative set contains $0, p-1$, proving (2.1).

Every residue has at most two representatives in $[0, 2p-2]$, so

$$|A + A| \le 2|\phi_c(V + V)| \le 2|V + V|.$$

Distinct residues in $\phi_c(W)$ give distinct integer differences, which proves (2.3). In $[-(p-1), p-1]$, residue zero has one representative and each nonzero residue has at most two. Hence

$$|A - A| \le 2|\phi_c(V - V)| - 1 \le 2|V - V| - 1 < 2p - 1.$$

This proves the lemma. □

3. The two-temperature theorem

Let $M \subset \mathbb{N}_0$ be finite, contain zero, and contain a positive integer. Define

$$S = M + M, \qquad D = M - M,$$

and, for $d \in D$,

$$\kappa(d) = \min\{a + b : a, b \in M,\ a - b = d\}.$$

For $x, y \in (0,1)$, put

$$P_-(x) = \sum_{d \in D} x^{\kappa(d)}, \qquad P_+(y) = \sum_{s \in S} y^s,$$

$$\mu = \frac{\sum_{d \in D} \kappa(d) x^{\kappa(d)}}{P_-(x)},$$

$$\rho = \log P_-(x) - \mu \log x, \qquad \sigma = \log P_+(y) - \mu \log y.$$

Theorem 3.1 (two-temperature cyclic quotient)

If $\sigma < \rho$, then

$$\boxed{C_{3a} \geq 2 - \frac{\sigma}{\rho}.} \tag{3.1}$$

More generally, for every real $r > \rho$,

$$\boxed{C_{3a} \geq 1 + \frac{\rho - \sigma}{r}.} \tag{3.2}$$

In particular, if a rational number $R > 1$ satisfies $\rho < \log R$, then

$$\boxed{C_{3a} \geq 1 + \frac{\rho - \sigma}{\log R}.} \tag{3.3}$$

Proof

Give D the symmetric Gibbs distribution

$$p_d = \frac{x^{\kappa(d)}}{P_-(x)}.$$

Its entropy is

$$H(p) = -\sum_d p_d \log p_d = \log P_-(x) - \mu \log x = \rho.$$

For every positive $d \in D$, set

$$n_d^{(n)} = n_{-d}^{(n)} = \lfloor n p_d \rfloor,$$

and put the unused count into zero:

$$n_0^{(n)} = n - 2 \sum_{d>0} \lfloor n p_d \rfloor.$$

These counts are nonnegative, sum to n, and their normalized frequencies converge to p. Moreover,

$$\sum_d n_d^{(n)} \kappa(d) = 2 \sum_{d>0} \lfloor np_d \rfloor \kappa(d) \le n\mu.$$

Let

$$L_n = \left\lfloor \frac{n\mu}{2} \right\rfloor,$$

$$V_n = \left\{ a \in M^n : \sum_i a_i \le L_n \right\},$$

and let $W_n \subset D^n$ be the type class with counts $n_d^{(n)}$.

Choose minimum-cost witnesses $\alpha_d, \beta_d \in M$ with

$$\alpha_d - \beta_d = d, \qquad \alpha_d + \beta_d = \kappa(d),$$

choosing the witnesses for $-d$ by swapping those for d. Every word in W_n has coordinate sum zero. Thus its total α-budget equals its total β-budget, while their sum is an even integer at most $n\mu$. Each budget is therefore at most L_n, and

$$W_n \subset V_n - V_n.$$

Writing $T_n = |W_n|$, Stirling's formula, or the standard method-of-types bounds on a fixed alphabet, gives

$$\frac{1}{n} \log T_n \longrightarrow H(p) = \rho. \tag{3.4}$$

Now use separate exponential tilts on the two sides. Every vector in $V_n + V_n$ has coordinates in S and total coordinate sum at most $2L_n$. Since $0 < y < 1$,

$$|V_n + V_n| y^{2L_n} \le P_+(y)^n.$$

Because $2L_n \le n\mu$,

$$|V_n + V_n| \le e^{n\sigma}. \tag{3.5}$$

Likewise, if $z = a - b \in V_n - V_n$, then $\kappa(z_i) \le a_i + b_i$ coordinatewise. Hence

$$|V_n - V_n| x^{2L_n} \le P_-(x)^n,$$

and

$$|V_n - V_n| \le e^{n\rho}. \tag{3.6}$$

Fix $r > \rho$. Put $N_n = \lceil e^{rn} \rceil$, and use Bertrand's postulate to choose a prime

$$N_n < p_n < 2N_n.$$

For all sufficiently large n, equations (3.4)-(3.6) give

$$p_n > |V_n - V_n|, \qquad p_n > 2\max M, \qquad T_n \geq 2.$$

Apply Lemma 2.1 to V_n, W_n, p_n. It gives a finite $A_n \subset [0, p_n - 1]$ satisfying the strict GHR hypothesis and

$$|A_n - A_n| > T_n/2, \qquad |A_n + A_n| \leq 2e^{n\sigma},$$

$$2\max A_n + 1 = 2p_n - 1.$$

Therefore

$$\theta(A_n) > 1 + \frac{\log T_n - n\sigma - \log 4}{\log(2p_n - 1)}.$$

By (3.4), the numerator after the initial 1, divided by n, tends to $\rho - \sigma$. The denominator divided by n tends to r. For every number strictly below the right side of (3.2), one sufficiently large fixed n therefore supplies a finite GHR seed. Applying the GHR lemma to that single seed proves (3.2). Letting $r \downarrow \rho$ proves (3.1), and putting $r = \log R$ gives (3.3). □

What is new relative to the one-temperature theorem

The earlier cyclic-quotient proof used $y = x$, so its numerator collapsed to $\log P_-(x) - \log P_+(x)$. The sum upper bound does not require this: its Markov parameter can be optimized independently. The new theorem uses the difference tilt x to choose the high-entropy realizable type and the sum tilt y only to upper-bound all possible sums at the resulting budget.

For the gapless limiting alphabet $\mathbb{N}_0$, taking $x = \sqrt{2} - 1$ and $y = 1/2$ gives

$$\rho = 2\log(1 + \sqrt{2}), \qquad \sigma = 2\log 2,$$

and therefore

$$2 - \frac{\sigma}{\rho} = 2 - \frac{\log 2}{\log(1 + \sqrt{2})} = 1.213560298642605\ldots.$$

Thus the independently announced $1.21356\ldots$ result of Logan Kleinwaks is the natural gapless baseline of the two-temperature theorem. The construction below crosses it by changing the inner alphabet, not by changing a cutoff in the old mask.

4. The structured alphabet $\langle 4, 5\rangle$

Let

$$H = \langle 4, 5\rangle.$$

It has conductor 12 and exactly six gaps:

$$\mathbb{N}_0 \setminus H = \{1, 2, 3, 6, 7, 11\}.$$

Take

$$M = H \cap [0, 80].$$

Exact reconstruction gives

$$|M| = 75, \qquad |M + M| = 155, \qquad |M - M| = 161.$$

In fact,

$$M + M = \{0, 4, 5, 8, 9, 10\} \cup [12, 160],$$

while

$$M - M = [-80, 80].$$

For every positive $d \in H \cap [1, 80]$, the witness $(d, 0)$ gives $\kappa(d) = d$. The six missing positive differences are recovered at the following exact costs:

d	1	2	3	6	7	11
$\kappa(d)$	9	18	13	14	17	19.

Thus the semigroup deletes six low-cost sum symbols but retains every difference symbol, paying a moderate witness-cost penalty only on those six gaps. That trade is exactly what improves the entropy ratio.

The partition polynomials can be written explicitly as

$$P_+(y) = 1 + y^4 + y^5 + y^8 + y^9 + y^{10} + \sum_{s=12}^{160} y^s,$$

and

$$P_-(x) = 1 + 2\left(\sum_{d \in H \cap [1,80]} x^d + x^9 + x^{18} + x^{13} + x^{14} + x^{17} + x^{19}\right).$$

5. Exact numerical certificate

Use

$$x = \frac{8149}{10000}, \qquad y = \frac{42571}{50000}, \qquad R = \frac{249293}{10000}.$$

The exact verifier reconstructs the polynomials and proves by integer cross-multiplication that

$$5.214722552042 < P_-(x) < 5.214722552043,$$

$$3.661102186601 < P_+(y) < 3.661102186602,$$

and

$$7.643532314898 < \mu < 7.643532314899.$$

It then uses rational atanh-series enclosures for every logarithm and proves

$$\log R - \rho > 0.000004277214540438787252509260 6, \tag{5.1}$$

and

$$\rho - \sigma - \frac{10709}{50000}\log R > 0.000002881889332672970012273824 7. \tag{5.2}$$

Equation (5.1) is the full-difference-rate condition in Theorem 3.1. Equation (5.2) gives

$$1 + \frac{\rho - \sigma}{\log R} > 1 + \frac{10709}{50000} = 1.21418.$$

Hence

$$\boxed{C_{3a} > 1.21418}.$$

Display-only values reconstructed from the exact integers are

$$\begin{aligned} P_-(x) &= 5.214722552042313553267783356 18\ldots, \\ P_+(y) &= 3.661102186601384133701906449 48\ldots, \\ \mu &= 7.643532314898500890118199465 50\ldots, \\ \rho &= 3.216039541306344061257102464 29\ldots, \\ \sigma &= 2.527224394365747114876546032 61\ldots, \\ \log R &= 3.216043818521118299072902158 84\ldots. \end{aligned}$$

6. One explicit finite GHR seed

Because x is rational, clear a common degree $D_0 = 160$. For each $d \in M - M$, put

$$\omega_d = 8149^{\kappa(d)} 10000^{160-\kappa(d)}.$$

Define

$$N = \sum_d \omega_d, \qquad 2L = \sum_d \kappa(d)\omega_d.$$

Then ω_d/N is exactly the Gibbs distribution, $2L$ is even, and

$$V = \left\{ a \in M^N : \sum_i a_i \le L \right\}.$$

Let W be the exact type class containing ω_d copies of each d. Its two witness budgets are exactly L, so $W \subset V - V$.

Set

$$T = \left\lfloor \left(\frac{249293}{10000} \right)^N \right\rfloor,$$

and let p be the least prime greater than T. Since $|V - V| < R^N$, we have $|V - V| \le T < p$. Bertrand's postulate gives $p < 2T < 2R^N$. Choose the quotient map by the canonical conditional- expectation rule from Lemma 2.1, rescale canonically so that a nonzero image element becomes -1, and let $A \subset [0, p-1]$ be the least- nonnegative representative set.

The standard finite type-class estimate is

$$|W| \ge \frac{e^{N\rho}}{(N+1)^{|M-M|}}.$$

Together with the quotient bounds and $2p - 1 < 4R^N$, a sufficient finite inequality for exponent greater than $1 + \alpha$, where $\alpha = 10709/50000$, is

$$N(\rho - \sigma - \alpha \log R) > |M - M| \log(N+1) + (2 + 2\alpha) \log 2. \tag{6.1}$$

The verifier proves (6.1) with rigorous rational intervals. Thus the result is not only a limit statement: the algorithm above specifies one finite GHR seed whose exponent exceeds 1.21418.

The exact integers have

$$\mathrm{bitlen}(N) = 2129, \qquad \mathrm{bitlen}(L) = 2131.$$

Their big-endian byte hashes are

```
N  a9db0611a20142dca241b32363fbb535cc64e9755d312f659edfe966add9e417
L  0f4a6eaef4e3f2387da12a5f1a0a2252c5cf929be806ee423a39957f137737e3
```

The complete decimal expansions are in the generated certificate directory. The prime, quotient coefficients, and all elements of A are finitely and canonically specified but are not materialized.

7. Verification boundary

The Python standard-library verifier performs:

1. exact reconstruction of $\langle 4, 5\rangle \cap [0, 80]$;
2. exact reconstruction of $M + M$, $M - M$, every $\kappa(d)$, and the partition-polynomial numerators;
3. exact rational enclosures for P_-, P_+, and μ;
4. rigorous rational enclosures for every logarithm;
5. the strict rate and exponent comparisons (5.1)-(5.2);
6. the explicit finite-size correction (6.1);
7. reconstruction and hashing of the exact finite N, L.

An independently written C++/GMP verifier uses Apéry shortest paths, intersection scanning for minimum witnesses, sparse monomial evaluation, and its own rational logarithm implementation. The two programs emit byte-for-byte identical masks, supports, difference-cost tables, and finite parameters.

The programs do **not** formally prove Theorem 3.1, the method-of-types bound, Bertrand's postulate, or the GHR finite-set lemma. Those remain written mathematical arguments in this note.

8. Reproduction

Python, standard library only:

```
python3 verification/verify_c3a_two_temperature_121418.py --emit certificate
```

Independent C++/GMP:

```
g++ -O2 -std=c++20 verification/verify_c3a_two_temperature_121418_gmp.cpp \
  -lgmpxx -lgmp -o verify_gmp
./verify_gmp --emit certificate_gmp
```

Then compare the five discrete certificate files byte for byte.

9. Acknowledgements and AI-use disclosure

I thank Logan Kleinwaks for reviewing an earlier `1.2060` candidate, identifying minor statement and exposition issues that are corrected here, and publicly sharing his independent `1.21356` construction. I also thank the contributors to the Optimization Constants in Mathematics project for maintaining the problem record and its reproducibility standards.

Carson Olafson used ChatGPT extensively to explore constructions, formulate the two-temperature argument, draft and revise this note, generate the verification programs, and audit the resulting package. Carson selected the research direction, ran and reviewed the verification outputs, and is the human contributor responsible for this submission. This note is neither externally refereed nor formally verified; independent review of Theorem 3.1 is expressly invited.

References

1. K. Gyarmati, F. Hennecart, and I. Z. Ruzsa, *Sums and differences of finite sets*, Functiones et Approximatio Commentarii Mathematici **37** (2007), 175–186, doi:10.7169/facm/1229618749.

2. L. Kleinwaks, review of an earlier cyclic-quotient construction and announcement of an independent `1.21356` bound, optimizationproblems PR #134 comment, August 15, 2026.

3. L. Kleinwaks, *Improve C_3a lower bound via Lean-formalized controlled-carry semigroup mask-digit construction*, optimizationproblems PR #146, August 14, 2026.